Jiangtian Li

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Academic Positions

Postdoctoral Fellow, Department of Linguistics, University of Toronto 2025–present
Advisor: Daphna Heller

Postdoctoral Fellow, Department of Psychology, University of Toronto Scarborough 2020–2025
Advisor: Blair Armstrong

Education

PhD in Philosophy, Western University 2016–2020
Advisor: Robert Stainton
Dissertation: On polysemy: A philosophical, psycholinguistic, and computational study

Research Associate, Department of Psychology & Brain and Mind Institute, Western University 2017–2020
Advisor: Marc Joanisse

MA in Philosophy, Western University 2015–2016
Advisor: Robert Stainton
Thesis: Names are also polysemous

BA in Philosophy, Minzu University of China 2009–2013
Advisor: Jixuan Zhang
Thesis: Truth and meaning: A defense of propositional deflationism

Publications

Journal Articles

Li, J., Wang, S., Ungerer, T., & Armstrong, B. C. (n.d.). PyLens: An easy to use Python-based neural network simulator for cognitive modeling. Under Review.

Li, J., Parker, M. S., Wang, Y., & Heller, D. (n.d.). Why modify? Examining mechanisms of redundant modification in referring expressions through the lens of state adjectives. Under review.

Li, J., Armstrong, B. C., & Xu, Y. (2026). Reply to Tong et al: Understanding the roles of continuous and categorical sense representations in child language acquisition. Accepted in *Proceedings of the National Academy of Sciences*

Li, J., Armstrong, B. C., & Xu, Y. (2026). Discovering regularity and mechanisms of word sense acquisition in childhood. *Proceedings of the National Academy of Sciences*, 123(20), e2525788123. https://doi.org/10.1073/pnas.2525788123

Li, J., & Armstrong, B. C. (2025). Issues of Generalization from Unreliable or Unrepresentative Stimuli: Broad Lessons from Lexical Ambiguity. *Open Mind*, 9, 1185–1250. https://doi.org/10.1162/opmi.a.21. [**Received the Best Paper Award in the Department of Psychology at the University of Toronto Scarborough.**]

Li, J. (2024). Semantic Minimalism and the Continuous Nature of Polysemy. *Mind & Language*, 1–26. https://doi.org/10.1111/mila.12509 [**Top 10 most-cited paper in *Mind & Language* among 2024 publications.**]

Li, J., & Armstrong, B. C. (2024). Probing the Representational Structure of Regular Polysemy via Sense Analogy Questions: Insights from Contextual Word Vectors. *Cognitive Science*, 48(3), e13416. https://doi.org/10.1111/cogs.13416

Li, J., & Joanisse, M. F. (2021). Word Senses as Clusters of Meaning Modulations: A Computational Model of Polysemy. *Cognitive Science*, 45(4), e12955. https://doi.org/10.1111/cogs.12955 [**Top cited article, *Cognitive Science*, 2021–2022.**]

Proceedings Articles

Cabana, A., Kaczer, L., Li, J., & Norman, R. (2026). New approaches to lexical ambiguity: Theory, methods, models, and measures. Proceedings of the Annual Meeting of the Cognitive Science Society, 48(0). https://escholarship.org/uc/item/1mc5m62z

Li, J., & Armstrong, B. C. (2024). Issues of Generalization from Unreliable or Unrepresentative Psycholinguistic Stimuli: A Case Study on Lexical Ambiguity. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46. https://escholarship.org/uc/item/52c2s25s (Among the top 19% of all submissions).

Li, J., & Armstrong, B. C. (2023). Probing the Representational Structure of Regular Polysemy in a Contextual Word Embedding Model via Sense Analogy Questions. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 45. https://escholarship.org/uc/item/300318x1 (Among the top 17.7% of all submissions).

Books

Li, J., & Armstrong, B. C. (2025). Lexical Ambiguity: An Interdisciplinary Exploration and Possible New Directions. Submitted for review at the University of Toronto Press.

Software

PyLens. A Python-based neural network simulator for cognitive modeling. Main code contributor, maintainer, and project supervisor. https://github.com/blairarmstrong/PyLensPublic.

Dissertation

Li, J. (2020). On Polysemy: A Philosophical, Psycholinguistic, and Computational Study. *Electronic Thesis and Dissertation Repository.* 7282. https://ir.lib.uwo.ca/etd/7282

Academic Presentations

Cabana, A., Kaczer, L., Li, J.*, & Norman, R. (2026). New approaches to lexical ambiguity: Theory, methods, models, and measures. Talk presented at CogSci 2026.

Li, J.*, Armstrong, B. C., & Xu, Y. (2026). Metasemantics and acquisition of word senses in children. Poster presented at the 52nd Annual Meeting of the Society for Philosophy and Psychology, Johns Hopkins University.

Li, J.*, Wang, Y., & Heller, D. (2026). From color to state: Production of redundant adjectives and limits of informativity-based models. Poster presented at the 2026 Annual Meeting of the Canadian Society for Brain, Behaviour and Cognitive Science, York University, Toronto.

Li, J.*, & Armstrong, B. C. (2026). Issues of generalization from unreliable or unrepresentative psycholinguistic stimuli: A case study on lexical ambiguity. Poster presented at the 2026 Annual Meeting of the Canadian Society for Brain, Behaviour and Cognitive Science, York University, Toronto.

Li, J.*, Armstrong, B. C., & Xu, Y. (2026). Discovering regularity and mechanisms of word sense acquisition in early childhood. Talk presented at the 2026 Annual Meeting of the Canadian Society for Brain, Behaviour and Cognitive Science, York University, Toronto.

Li, J., Wang, Y., & Heller, D.* (2026). From color to state: Overspecification and the limits of informativity-based models. Poster presented at the 39th Annual Conference on Human Sentence Processing, Boston, USA.

Li, J.*, Armstrong, B. C., & Xu, Y. (2025). Discovering regularity and mechanisms of word sense acquisition through time. Talk presented at the UCL–UofT Workshop on Word Meaning and Ambiguity, London, UK.

Li, J.*, Armstrong, B. C., & Xu, Y. (2025). Discovering regularity in word sense acquisition through time. Talk presented at the 55th *Annual Meeting of the Society for Computation in Psychology*, Denver, USA.

Li, J.*, Armstrong, B. C., & Xu, Y. (2025). Discovering regularity in word sense acquisition through time. Talk presented at the UCL–UofT Workshop on Word Meaning and Ambiguity, Toronto, Canada.

Armstrong, B. C.*, & Li, J. (2025). Neural network model insights into lexical ambiguity. Invited symposium presentation at the Experimental Psychology Society (UK), Lancaster, UK.

Li, J.* (2024). Issues of generalization from unreliable or unrepresentative psycholinguistic stimuli: A case study on lexical ambiguity. Talk presented at the *Cognitive Science of Language (CSoL)* research group, University of Toronto.

Li, J.*, & Armstrong, B. C. (2024). Issues of generalization from unreliable or unrepresentative psycholinguistic stimuli: A case study on lexical ambiguity. Talk presented at the *Annual Meeting of the Cognitive Science Society*, Rotterdam, Netherlands (Among the top 19% of all submissions).

Li, J.*, & Armstrong, B. C. (2023). Probing the representational structure of regular polysemy in a contextual word embedding model via sense analogy questions. Talk presented at the *Annual Meeting of the Cognitive Science Society*, Sydney, Australia (Among the top 17.7% of all submissions).

Li, J.* (2021). Polysemy as clusters of meaning modulations. Talk presented at the *Language and Cognition Lab*, UC San Diego.

Li, J.* (2020). On polysemy: A philosophical, psycholinguistic, and computational study. Talk presented at the *Psycholinguistics* research group, University of Toronto.

Journal and Conference Reviews

Psychological Review, Psychonomic Bulletin & Review, Mind & Language, Language Learning, Behavior Research Methods, Cognitive Science, Glossa Psycholinguistics, Applied Psycholinguistics, Canadian Journal of Experimental Psychology, Quarterly Journal of Experimental Psychology, Language, Cognition and Neuroscience, Natural Language Processing (Cambridge), Humanities & Social Sciences Communications,

Annual Meeting of the Cognitive Science Society

Grant Reviews

National Science Foundation 2022

Grants, Honors & Awards

University of Toronto Scarborough Psychology Best Paper Award ($250) 2026
Western Graduate Research Scholarship (≈ $18k each year) 2015–2020
Chair's Entrance Scholarship ($1000) 2015
Ranked 1st in Philosophy Summer School in China and awarded fellowship (£3000) 2014
Secondary Professional Scholarship 2013
Hong Kong Xinshan Scholarship 2010

Academic Activities

Member of the Committee of Session & Technical Chairs of CogSci 2022 2022
Graduate Research Assistant at the University of Western Ontario 2020
Commenter at UWO Graduate Conference in Philosophy of Mind, Language, and Cognitive Science 2018
Commenter at UWO Graduate Conference in Philosophy of Mind, Language, and Cognitive Science 2017
Attended the North American Summer School on Logic, Language, and Information (NASSLLI) 2016
Assistant Editor in Philosophy of Language, PhilPapers 2015–2016

Course Instruction

Introduction to Philosophy of Mind, University of Western Ontario 2019

Guest Lecturing

Computers in Linguistics, University of Toronto 2023
Philosophy of Mind, University of Western Ontario 2019
Introduction to Philosophy of Language, University of Western Ontario 2017

Teaching Assistantships

Philosophical Writing and Methodology, University of Toronto 2024
Philosophy of Neuroscience, University of Western Ontario 2019
Understanding Science, University of Western Ontario 2018
Critical Thinking, University of Western Ontario 2017–2018
Introduction to Philosophy of Language, University of Western Ontario 2017
Understanding Science, University of Western Ontario 2016

Languages and Technical Skills

Languages: English (fluent), Mandarin (native), Japanese (fluent)
Technical Skills: neural network modeling, Bayesian probabilistic modeling, statistical analysis with mixed-

effects models, natural language processing, experimental design, corpus analysis